Finite-depth Quasi-Phase Matching (QPM) and Converter Efficiency

Abhishek V. Karve
Department of Applied Physics, Stanford University
(Dated: October 15, 2026)

An ensemble of ^{85}Rb atoms couples to optical (780 nm) and microwave cavities. A 480/481 nm blue control connects optical and Rydberg excitations; 297 nm UV dressing enables the effective microwave coupling from the prepared atomic state, completing the conversion loop [2]. A separate 1188 nm lattice confines the atoms and provides the periodic density for quasi-phase matching (QPM).

Finite site width reduces the coherent overlap Q between the two atomic conversion paths. This note derives the resulting efficiency, including cavity input and output coupling, and gives a numerical example. The derivation is self-contained; the companion Mathematica notebook implements and checks it, with a notation guide at the end [3].

I. WHAT THE FINITE-DEPTH QPM FACTOR MEANS

For atoms $j = 1, \ldots, N$, let u_j, v_j be the complex excitation amplitudes in the normalized optical/blue and UV/microwave collective profiles. They include relative coupling magnitudes and phases; overall strengths enter separately in section II. Define

$$Q = \left| \sum_j u_j^* v_j \right|, \qquad \sum_j |u_j|^2 = \sum_j |v_j|^2 = 1, \qquad 0 \le Q \le 1. \tag{1}$$

For matched, uniform envelopes, $Q = |\langle e^{i\Delta\mathbf{k}\cdot\mathbf{r}}\rangle|$, averaged over the normalized atomic spatial distribution, with $\Delta\mathbf{k} = \mathbf{k}_{780} + \mathbf{k}_{480} - \mathbf{k}_{\mathrm{UV}} - \mathbf{k}_{\mu}$. Wavevectors include propagation directions; reversing the loop reverses $\Delta\mathbf{k}$ without changing Q. Thus Q is an amplitude overlap and Q^2 the squared useful-mode projection. Conversion probability also depends on coherent exchange, atomic decay, and cavity ports. Neither Q nor Q^2 alone is a conversion probability.

A. Localization within one site.

An ideal retroreflected lattice has period $d = \lambda_{\mathrm{L}}/2$ and reciprocal wavevector $K = 2k_{\mathrm{L}}$, where $k_{\mathrm{L}} = 2\pi/\lambda_{\mathrm{L}}$. For displacement x from a site minimum, barrier height U, and atomic mass m, the harmonic approximation gives the recoil energy E_{R} and trap angular frequency ω_t [1]:

$$V(x) = U \sin^2(k_{\mathrm{L}} x) \simeq U k_{\mathrm{L}}^2 x^2, \quad E_{\mathrm{R}} = \frac{\hbar^2 k_{\mathrm{L}}^2}{2m}, \quad \omega_t = \frac{2\sqrt{U E_{\mathrm{R}}}}{\hbar}. \tag{2}$$

With reduced Planck constant $\hbar$, Boltzmann constant k_{B}, and temperature T, the thermal position variance and density Fourier harmonics are

$$\begin{aligned} \sigma_x^2 &= \frac{\hbar}{2m\omega_t} \coth\!\left(\frac{\hbar\omega_t}{2k_{\mathrm{B}}T}\right), \\ b_n &= \left|\left\langle e^{inKx}\right\rangle\right| = e^{-(nK)^2\sigma_x^2/2} = \exp\!\left[-n^2\sqrt{\frac{E_{\mathrm{R}}}{U}}\coth\!\left(\frac{\sqrt{U E_{\mathrm{R}}}}{k_{\mathrm{B}}T}\right)\right]. \end{aligned} \tag{3}$$

The integer n is a spatial harmonic order, not a vibrational band. At $T = 0$, coth $\to 1$, but zero-point motion keeps $b_n < 1$ for finite U and $n \neq 0$. Deeper trapping drives $b_n \to 1$ at fixed T. This approximation requires localization near the minimum; occupation near the barrier requires band wavefunctions.

B. From one axis to the total overlap.

Decompose position into the site centre $\mathbf{R}$ and the in-site displacement. Exact matching, $\Delta\mathbf{k} = \sum_i n_i \mathbf{K}_i$ with integer n_i, makes equivalent site centres contribute in phase. Finite in-site widths still reduce the overlap; for independent

confined directions,

$$Q = \prod_i b_{n_i,i}; \qquad Q = b_{2,x} b_{2,y} \quad \text{for two second-order QPM axes.} \tag{4}$$

The last expression is the working geometry used below: co-propagating 780 and 480/481 nm fields, perpendicular UV, and suitable lattice axes. It is a *conditional geometry*, not fixed by the nominal wavelengths alone. A residual mismatch $\delta\mathbf{k}$ also produces phase cancellation across occupied site centres, through $|\langle e^{i\delta\mathbf{k}\cdot\mathbf{R}}\rangle|$. Beam angles, exact wavelengths, and mode envelopes must therefore be checked separately.

II. ANALYTICAL CONVERTER MODEL WITH THE ORTHOGONAL MODES

A. Problem definition and collective-mode construction.

The system contains an optical cavity mode $\hat{a}$, a microwave cavity mode $\hat{b}$, and an ensemble with optically excited and Rydberg excitations. The optical cavity couples to one collective atomic profile with strength g_o; the UV-dressed microwave interaction couples to another with strength g_m. The blue field connects the optical and Rydberg excitations with matrix element Ω. The profiles need not coincide: their normalized overlap is Q from Eq. (1).

We drive only the optical input and seek the coherent microwave output in the linear, weak-excitation regime. Efficiency is the ratio of output microwave photon flux to input optical photon flux; control fields supply the photon-energy difference. This is a photon-number efficiency, not an energy-efficiency ratio. All analytical couplings, detunings, and linewidths are angular frequencies; section III gives numerical values divided by 2π.

1. *Atomic variables and assumptions.*

Let $\hat{e}_j^\dagger$ and $\hat{r}_j^\dagger$ create optical and Rydberg excitations on atom j relative to the prepared reference state. The optical excitation is a ground-to-optically-excited-state coherence; the Rydberg excitation is the effective coherence addressed by the blue and UV-dressed microwave paths. When signal excitations are few compared with the prepared atoms, depletion and saturation are negligible and these operators are approximately bosonic:

$$[\hat{e}_j, \hat{e}_\ell^\dagger] \simeq [\hat{r}_j, \hat{r}_\ell^\dagger] \simeq \delta_{j\ell}, \qquad [\hat{e}_j, \hat{r}_\ell^\dagger] \simeq 0. \tag{5}$$

Assume frozen positions, uniform blue-control magnitude, identical atomic detunings and independent decay rates. UV dressing is included in g_m and its profile. Absorbing the blue phase into $\hat{r}_j$ makes Ω real and uniform. It is the Hamiltonian matrix element below; convert any Rabi-frequency convention containing an extra factor of $1/2$ before substitution:

$$\frac{\hat{H}_B}{\hbar} = -\Omega \sum_j \left(\hat{e}_j^\dagger \hat{r}_j + \hat{r}_j^\dagger \hat{e}_j\right). \tag{6}$$

2. *The two coupling profiles.*

In this phase convention, the optical cavity selects $\mathbf{u}$ and the UV-dressed microwave interaction selects $\mathbf{v}$, as in section I:

$$\mathbf{u}^\dagger\mathbf{u} = \mathbf{v}^\dagger\mathbf{v} = 1, \qquad \mathcal{Q} = \mathbf{u}^\dagger\mathbf{v} = \sum_j u_j^* v_j, \qquad Q = |\mathcal{Q}|. \tag{7}$$

Absorb the global phase of $\mathbf{v}$ into $\hat{b}$ to choose $\mathbf{u}^\dagger\mathbf{v} = Q \geq 0$. With $s = \sqrt{1-Q^2}$,

$$\boxed{\mathbf{v} = Q\mathbf{u} + s\mathbf{w},} \qquad \mathbf{w} = \frac{\mathbf{v} - Q\mathbf{u}}{s}, \qquad \mathbf{u}^\dagger\mathbf{w} = 0, \quad \mathbf{w}^\dagger\mathbf{w} = 1. \tag{8}$$

The formula for $\mathbf{w}$ applies for $Q < 1$. At $Q = 1$, any normalized orthogonal profile suffices because its microwave coupling vanishes ($s = 0$).

3. Collective operators and closure.

Define four collective operators (uppercase letters distinguish operators from their mean amplitudes):

$$\begin{aligned} \hat{P}^\dagger &= \sum_j u_j \hat{e}_j^\dagger, \quad \hat{R}^\dagger = \sum_j u_j \hat{r}_j^\dagger, \\ \hat{P}_\perp^\dagger &= \sum_j w_j \hat{e}_j^\dagger, \quad \hat{R}_\perp^\dagger = \sum_j w_j \hat{r}_j^\dagger. \end{aligned} \tag{9}$$

In the weak-excitation limit, orthonormality gives unit commutators within each mode; distinct modes commute. The coupling chain is

$$a \overset{g_o}{\longleftrightarrow} E_1 \overset{\Omega_B}{\longleftrightarrow} R_1 \overset{Qg_m}{\longleftrightarrow} b \overset{sg_m}{\longleftrightarrow} R_2 \overset{\Omega_B}{\longleftrightarrow} E_2. \tag{10}$$

Here a, b label the optical and microwave cavity modes. The chain labels E_1, R_1, R_2, E_2 represent $\hat{P}, \hat{R}, \hat{R}_\perp, \hat{P}_\perp$, respectively, and $\Omega_B = \Omega$. The two cavity coupling profiles span at most a two-dimensional atomic subspace, preserved by uniform control, detunings, and damping. Consequently, one additional optical/Rydberg pair suffices: the response closes on four atomic and two cavity modes. All other orthogonal atomic modes are initially unexcited and have zero mean drive.

4. Hamiltonian and conventions.

The collective matrix elements g_o, g_m include each profile's total coupling magnitude and are held fixed as Q varies. They are collective couplings, not single-atom matrix elements. The baseline parameter convention is [3]

$$g_o = \frac{G}{\sqrt{2}}, \qquad g_m = \frac{G_\mu}{\sqrt{2}}. \tag{11}$$

Thus G, G_μ rescale the same collective couplings. With zero optical-cavity detuning, the rotating-frame Hamiltonian fixes the signs of the optical atomic, Rydberg, and microwave detunings $\Delta_e, \Delta_r, \Delta_\mu$:

$$\frac{\hat{H}_0}{\hbar} = -\Delta_\mu \hat{b}^\dagger \hat{b} - \Delta_e(\hat{P}^\dagger \hat{P} + \hat{P}_\perp^\dagger \hat{P}_\perp) - \Delta_r(\hat{R}^\dagger \hat{R} + \hat{R}_\perp^\dagger \hat{R}_\perp), \tag{12}$$

$$\frac{\hat{H}_{\rm int}}{\hbar} = -\Big[g_o \hat{a}^\dagger \hat{P} + \Omega(\hat{P}^\dagger \hat{R} + \hat{P}_\perp^\dagger \hat{R}_\perp) + g_m \hat{b}^\dagger (Q\hat{R} + s\hat{R}_\perp) + \text{h.c.}\Big]. \tag{13}$$

Use $\dot{\hat{c}} = (i/\hbar)[\hat{H}, \hat{c}]$ and Fourier amplitudes $e^{+i\omega t}$, with spectral offset ω. For optical input and useful microwave output coupling rates $\kappa_{o,\rm ex}, \kappa_{m,\rm ex}$, the optical drive is

$$\frac{\hat{H}_{\rm d}}{\hbar} = -\sqrt{\kappa_{o,\rm ex}}\left[\hat{a}^\dagger a_{\rm in}(t) + a_{\rm in}^*(t)\hat{a}\right]. \tag{14}$$

Here $|a_{\rm in}|^2$ is incident photon flux. Write $a = \langle\hat{a}\rangle$, $b = \langle\hat{b}\rangle$, $p = \langle\hat{P}\rangle$, $r = \langle\hat{R}\rangle$, $p_\perp = \langle\hat{P}_\perp\rangle$, and $r_\perp = \langle\hat{R}_\perp\rangle$. With no microwave input, $b_{\rm out} = \sqrt{\kappa_{m,\rm ex}}\, b$ up to a port phase, and $S_{mo} = b_{\rm out}/a_{\rm in}$. Reservoir noise has zero mean; this response calculation does not characterize added noise.

B. Response functions and linewidth conventions.

Let κ_o, κ_m be the total cavity energy decay rates, including all ports and internal loss. Thus $\kappa_{o,\rm ex} \le \kappa_o$ and $\kappa_{m,\rm ex} \le \kappa_m$. The atomic linewidth parameters Γ, γ_r give optical and Rydberg amplitude decay rates $\Gamma/2, \gamma_r/2$, respectively.

The baseline equations additionally assign microwave amplitude damping $\gamma_r/2$, giving effective full width $\kappa_m + \gamma_r$ [3]. This is a separate phenomenological assumption, not a consequence of QPM or a change to the useful port rate. Including damping, the inverse response functions are

$$\begin{aligned} D_a &= \kappa_o/2 + i\omega, & D_e &= \Gamma/2 + i(\omega - \Delta_e), \\ D_r &= \gamma_r/2 + i(\omega - \Delta_r), & D_b &= (\kappa_m + \gamma_r)/2 + i(\omega - \Delta_\mu). \end{aligned} \tag{15}$$

Real parts give amplitude damping; imaginary parts give detuning. Orthogonal-mode feedback is derived separately below.

C. Six coupled equations.

With these conventions,

$$\begin{aligned} D_a a - i g_o p &= i\sqrt{\kappa_{o,\mathrm{ex}}}\, a_{\mathrm{in}}, \\ D_e p - i g_o a - i\Omega r &= 0, \\ D_r r - i\Omega p - i g_m Q b &= 0, \\ D_b b - i g_m \big(Q r + \sqrt{1-Q^2}\, r_\perp\big) &= 0, \\ D_r r_\perp - i\Omega p_\perp - i g_m \sqrt{1-Q^2}\, b &= 0, \\ D_e p_\perp - i\Omega r_\perp &= 0. \end{aligned} \tag{16}$$

The microwave coupling splits into useful amplitude Qg_m and orthogonal amplitude $g_m\sqrt{1-Q^2}$, retaining total strength g_m.

D. Eliminating the orthogonal pair.

Substitute $p_\perp = i\Omega r_\perp/D_e$ from the last equation into the preceding one. With $F = D_e D_r + \Omega^2$,

$$r_\perp = \frac{i g_m \sqrt{1-Q^2}\, D_e}{F}\, b, \qquad \boxed{\Sigma_\perp(\omega) = \frac{g_m^2 (1-Q^2) D_e}{F}}. \tag{17}$$

The microwave inverse response becomes $D_b + \Sigma_\perp$. The eliminated pair adds amplitude damping $\mathrm{Re}\,\Sigma_\perp$ (full linewidth $2\,\mathrm{Re}\,\Sigma_\perp$) and dispersion $\mathrm{Im}\,\Sigma_\perp$. Despite lacking direct optical-cavity coupling, it is microwave-driven, feeds back on that field, and dissipates through atomic decay. Replacing G_μ by QG_μ alone discards this feedback. Neither this damping nor the baseline extra damping belongs in the useful output-port rate.

E. Closed transfer function.

With $B = D_b + \Sigma_\perp$, eliminating the remaining atomic amplitudes gives

$$S_{mo} = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, g_o g_m \Omega Q}{B(D_a F + g_o^2 D_r) + g_m^2 Q^2 (D_a D_e + g_o^2)}. \tag{18}$$

Set $Y = D_a F + g_o^2 D_r$, $Z = D_b F + g_m^2 D_e$. Substituting $\Sigma_\perp$ and $F = D_e D_r + \Omega^2$ gives, up to the port phase,

$$\boxed{S_{mo}(\omega) = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, g_o g_m \Omega Q F}{YZ + Q^2 g_o^2 g_m^2 \Omega^2}, \qquad \eta(\omega) = |S_{mo}(\omega)|^2.} \tag{19}$$

Because Q enters both numerator and denominator, efficiency cannot generally be scaled by Q^2. At $Q = 1$ the orthogonal pair decouples, recovering the four-mode baseline; at $Q = 0$ coherent conversion vanishes.

III. ON-RESONANCE EFFICIENCY AND THE NUMERICAL VALUE OF Q

At $\omega = \Delta_e = \Delta_r = \Delta_\mu = 0$, define cooperativities comparing coherent coupling to damping:

$$C_o = \frac{2G^2}{\kappa_o \Gamma}, \qquad C_m = \frac{2G_\mu^2}{(\kappa_m + \gamma_r)\gamma_r}, \qquad C_B = \frac{4\Omega^2}{\Gamma \gamma_r}. \tag{20}$$

Let $\eta_o = \kappa_{o,\mathrm{ex}}/\kappa_o$ and $\eta_m = \kappa_{m,\mathrm{ex}}/\kappa_m$. Equation (19) becomes

$$\boxed{\eta(0) = \eta_o \eta_m \frac{\kappa_m}{\kappa_m + \gamma_r} \frac{4Q^2 C_o C_m C_B (1+C_B)^2}{\big[(1+C_B+C_o)(1+C_B+C_m) + Q^2 C_o C_m C_B\big]^2}.} \tag{21}$$

Port fractions multiply the efficiency at fixed total linewidths. Changes to mirror or microwave-port coupling that alter linewidths require updated cooperativities. Propagation and detection transmissions further multiply this port-to-port result to give end-to-end efficiency.

A. Lattice example.

Take ^{85}Rb, $\lambda_{\mathrm{L}} = 1188\,\mathrm{nm}$, waist $w = 200\,\mu\mathrm{m}$, and 100% retroreflection. Adopt the specified calibration 3 W per beam per axis $\leftrightarrow U/k_{\mathrm{B}} = 15\,\mu\mathrm{K}$; no polarizability calculation is implied. With $E_{\mathrm{R}}/k_{\mathrm{B}} = 0.0799004\,\mu\mathrm{K}$, $T = 0$, and two equal, exactly matched second-order axes,

$$b_2 = e^{-4\sqrt{E_{\mathrm{R}}/U}} = 0.746816, \qquad \boxed{Q = b_2^2 = 0.557734}, \qquad Q^2 = 0.311067. \tag{22}$$

This harmonic lowest-band overlap includes zero-point motion but does not resolve motional sidebands. It enters the converter as a static spatial overlap.

B. Converter example.

Use the collective couplings and linewidths

$$\begin{aligned} \frac{(G, G_\mu, \Omega)}{2\pi} &= (15.8474565,\ 2.09170145,\ 5.52497339)\,\mathrm{MHz}, \\ \frac{(\kappa_o, \kappa_m, \Gamma, \gamma_r)}{2\pi} &= (10,\ 5,\ 6,\ 0.1)\,\mathrm{MHz}. \end{aligned} \tag{23}$$

All detunings are zero. Consistently dividing every rate and frequency by 2π leaves the efficiency unchanged, allowing direct use of these MHz values. For unit port fractions and the finite-depth Q of Eq. (22):

Model	On-resonance efficiency
Matched profiles, $Q = 1$	82.0957%
Only $G_\mu \mapsto QG_\mu$; orthogonal pair omitted	51.5929%
Six modes with $Q = 0.557734$	**47.0824**%

Here $\Sigma_\perp(0)/(2\pi) = 0.147394\,\mathrm{MHz}$ and the additional full linewidth is $2\,\mathrm{Re}\,\Sigma_\perp(0)/(2\pi) = 0.294787\,\mathrm{MHz}$. Port fractions $\eta_o = \eta_m = 0.98$ give 45.21798% at unchanged total linewidths, before further path losses.

C. Scope and checks.

The closure requires weak excitation, frozen positions, uniform control and atomic linewidths, and one effective optical profile. Anharmonic motion, resolved sidebands, differential light shifts, or further independent standing-wave channels require an extended susceptibility. Mathematica checks reproduce the six-equation solution and $Q = 1$ limit. Numerical predictions remain conditional on the stated geometry.

D. Using the companion Mathematica notebook.

`Abhishek_Converter_QPM.nb` solves the six equations, eliminates the orthogonal pair, checks $Q = 1$, and evaluates the example. Its κ, κ_μ are κ_o, κ_m here; other physical symbols are unchanged. The calculator's $G_{\mathrm{opt}}, G_{\mathrm{mm}}$ are G, G_μ. Set the auxiliary parameter $\epsilon = 1$.

The notebook's forcing parameter `Ein` remains in its amplitude `ConverterEfficQ` $= \sqrt{\kappa_o\kappa_m}\,b$. The input-normalized response, up to the port phase, is

$$S_{mo} = \sqrt{\eta_o\eta_m}\,\frac{\texttt{ConverterEfficQ}}{\texttt{Ein}}, \qquad \eta = \eta_o\eta_m\,\texttt{NormalizedEfficiencyQ}, \tag{24}$$

Here $\epsilon = 1$ and `NormalizedEfficiencyQ` $=$ `EfficiencyQ`/`Ein`^{2} for real `Ein`. Examples set `Ein`=1; normalized linear efficiency is independent of drive amplitude.

`SigmaPerp` implements Eq. (17); `OnResonanceEfficiencyQ` implements Eq. (21) with unit port fractions.

`parmsScreenshot` gives Eq. (23), while `parmsOriginal` retains the original notebook's different parameters. Keep Q symbolic until applying `qRule`. The notebook's x, z labels denote the same two matched axes called x, y here. Its `etaOpt`,`etaMM` are η_o, η_m, applied by `EfficiencyQWithPorts` at fixed linewidths.

---

[1] I. Bloch, J. Dalibard and W. Zwerger, *Rev. Mod. Phys.* **80**, 885 (2008), Sec. II. doi:10.1103/RevModPhys.80.885.

[2] A. Kumar et al., *Nature* **615**, 614–619 (2023). doi:10.1038/s41586-023-05740-2.

[3] Baseline model: `Abhishek_Converter.nb`; companion implementation: `Abhishek_Converter_QPM.nb`. Calculator: AVKGitHub1/transduction_calcs, commit `30da8841b3c9`. The Q extension is derived here.